Senegal Seaports, Hydro Waste Waterways, Water Stress, Plastic Pollution in Relation to Population Density with Water Reservoirs

Author: Meya Johnson , Advisor : Catherine DuBreck

Rochester Institute of Technology, GIS/SIG Association , New York Water Environment Association Inc.

Research Question

How do spatial patterns of population density, industrial and urban wastewater discharge, and hydrological connectivity influence the distribution and seasonal dynamics of marine plastic pollution in Senegal's coastal zones, and how can integrated GIS-based modeling inform targeted interventions aligned with SDGs 6, 11, 12, and 14 to optimize waste management and marine ecosystem protection?"

Literature Review

SDG 6 (Clean Water and Sanitation): Marine plastic pollution directly affects water quality by contaminating drinking water sources, clogging drainage systems, and exacerbating sanitation challenges. In Dakar, uncollected plastic waste accumulates in drainage channels, increasing the risk of waterborne diseases and reducing access to clean water. GIS mapping of plastic waste hotspots can help authorities improve waste management strategies, ensuring cleaner water bodies and enhanced sanitation infrastructure.

The application of SDG 6 in mapping efforts highlights the urgent need to address pollution sources to safeguard public health and environmental sustainability. SDG 11 (Sustainable Cities and Communities): Addressing plastic pollution is crucial for creating cleaner and more sustainable urban environments, particularly in high-density areas like Dakar. With a growing population and rapid urbanization, the city faces increasing waste management challenges.

Methods and Data Resources

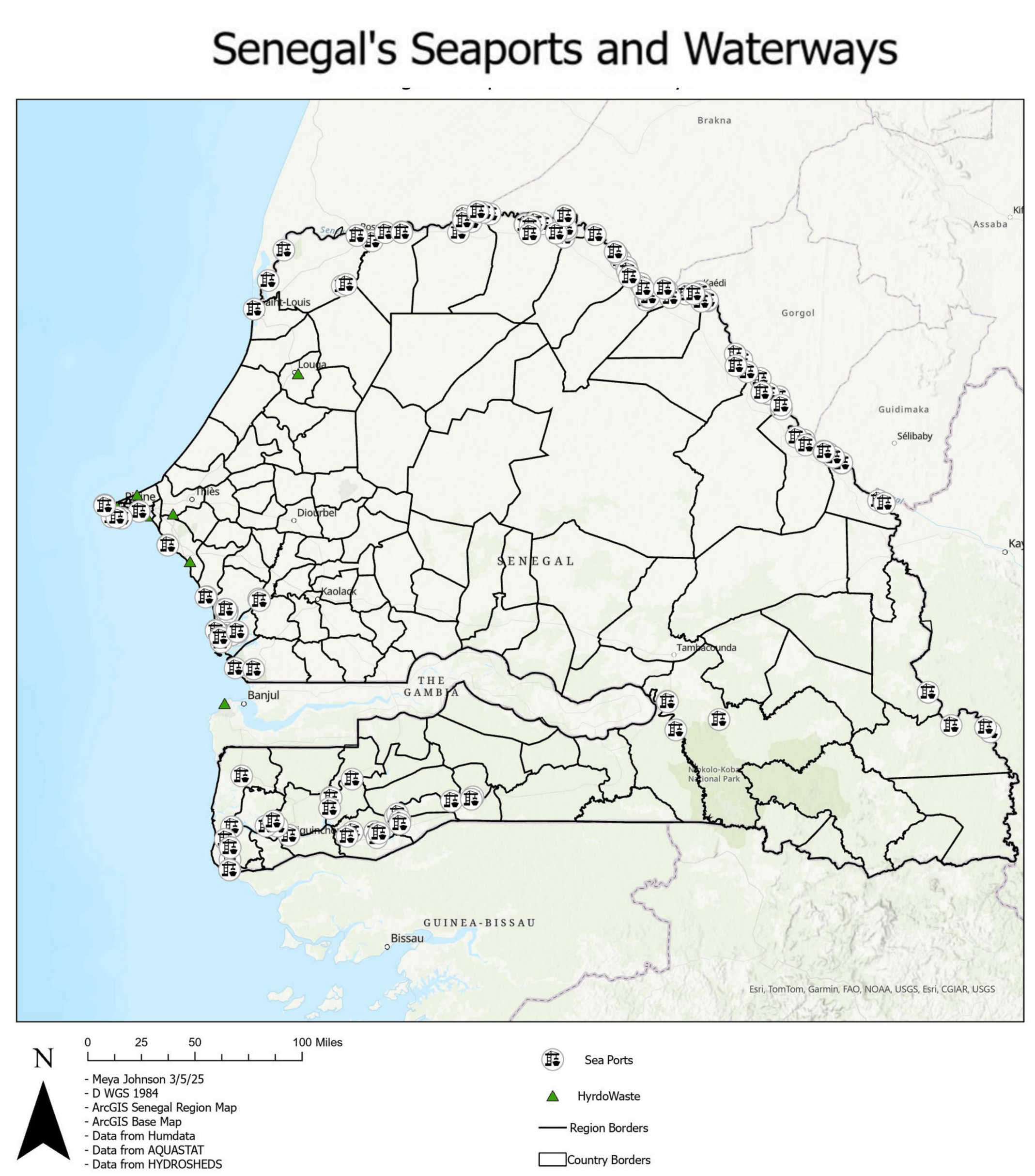
Possible methods entails utilizing satellite imagery and remote sensing to identify locations where plastic waste accumulates. By examining spatial data on plastic concentration along coastlines and rivers, policymakers can identify priority zones for waste collection and intervention. Furthermore, GIS-based flood risk mapping can forecast how plastic waste moves through waterways during heavy rains. By combining hydrological data with waste distribution models, authorities can create focused waste management strategies, such as placing waste traps in areas at high risk before floods carry plastic into the ocean.

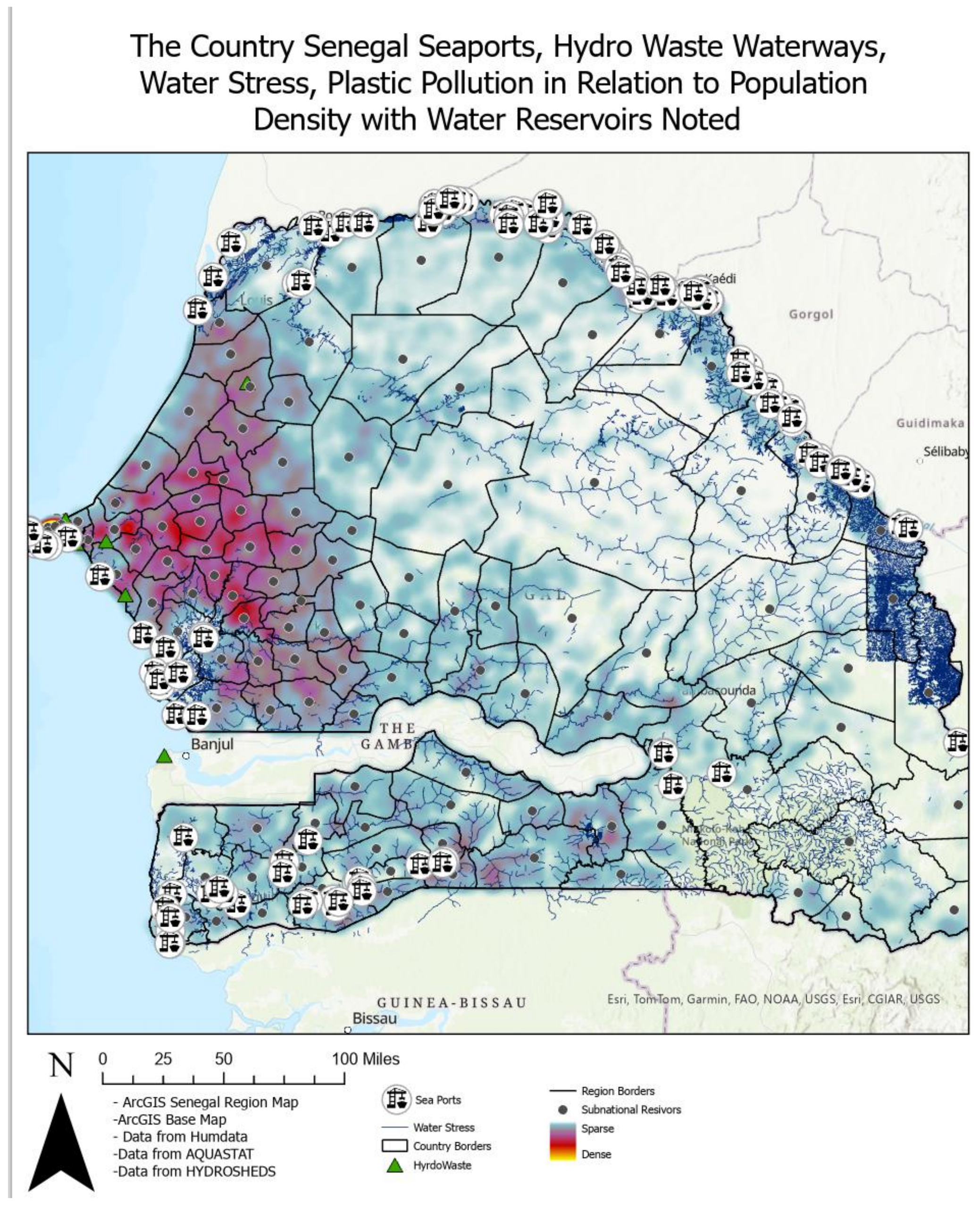
Station name	D_{efflu} (m)	D_{coast} (m)	Depth (m)	$D_{micropl}$ (m)	Area $_{micropl}$ (m^2)	GPS Lat. North, Long. West	Type of pollution	Effluent flow	Nature of the bottom
Cambérène	474	447	7.3	769	916	14.771338, −17.431724	Wastewater plant effluent	H	S
Yoff Tonghor	143	122	2.2	630	725	14.766937, −17.478672	Wastewater Fishing landing place	M	R

The expected advantages of implementing GIS solutions comprise:
Enhanced waste management strategies driven by spatial data insights.
Improved tracking of ghost fishing, which has a significant impact on Senegalese waters and spreads across the country through GPS monitoring.
More effective monitoring and reduction of plastic pollution in critical regions. Strengthened flood prevention measures to diminish plastic waste transportation into marine environments. Increased public awareness through visual mapping tools that emphasize pollution hotspots.

Senegal's Stressors on the Senegalese Environment from the Plastic pollution

Urbanization is intensifying the global resource and waste crisis. In lower-income nations, waste generation is projected to more than double over the next twenty years. Conventional waste collection and unregulated disposal methods, which do not include resource recovery components, are increasingly seen as economically, environmentally, and climatically unsustainable. An estimated 80% of wastewater globally is released untreated, and open defecation continues to be widespread in various regions. These practices create significant health, safety, and environmental dangers, especially for marginalized urban communities [2]. GIS technology can play a crucial role in reducing marine plastic pollution in Senegal. Sustainable tourism policies in Senegal emphasize conservation and resource management, as reflected in the geographic trends of high population to high plastic population density seen outlined in Map 1.





Background

Senegal, a West African nation with a coastline stretching over 700 kilometers along the Atlantic Ocean, faces significant challenges related to marine plastic pollution. As a rapidly developing country with major coastal cities such as Dakar, Saint-Louis, and Mbour, Senegal relies heavily on its fishing and tourism industries. However, plastic waste accumulation in waterways and coastal zones threatens marine biodiversity, disrupts local economies, and poses health risks to communities. Senegal also suffers from a chronic deficit of access to solid waste management services. Senegal produces over 2.4 million tons of solid waste per year, out of which 1.08 million tons is uncollected. The collection rate in Senegal on average is 55%, below the average in sub-Saharan Africa (65%) [3], [4]. As the hub of economic and industrial activities, Dakar is the main "solid waste producer" in Senegal, but solid waste management practices are not yet aligned with the heavy amount of waste generated [5]. Senegal's marine plastic pollution issue is aggravated by insufficient waste management infrastructure, high population density in urban areas, and increasing plastic consumption. Plastic waste often enters the environment through open dumpsites, stormwater runoff, and direct disposal into rivers and oceans. Seasonal flooding worsens the problem by carrying plastic debris into the Atlantic, further contributing to ocean pollution.

Picture 1: Compilation of pictures of inland and marine life pollution in Dakar, Senegal [(Roy), (UNEP)]

Conclusion

GIS-driven solutions can inform national policy, urban development planning, and grassroots cleanup initiatives, ensuring that Senegal's coastal and marine ecosystems remain viable for future generations. On a broader scale, addressing marine plastic pollution aligns with key Sustainable Development Goals (SDGs). Senegal, for instance, has launched public campaigns aimed at reducing plastic waste and promoting alternative materials to curb reliance on single-use plastics. The objective is to educate communities about the dangers of plastic pollution and encourage sustainable consumption habits. Measures include promoting reusable products and recycling programs to mitigate marine plastic pollution. Sustainable tourism policies in Senegal emphasize conservation and responsible resource management.

The goal is to balance economic growth with environmental protection, leveraging tourism as a tool for biodiversity conservation. Efforts to combat marine plastic pollution focus on safeguarding coastal and marine environments, which are vital for biodiversity and the livelihoods of coastal populations. Senegal's approach includes regional collaboration, improved waste management systems, and widespread public awareness campaigns to combat the escalating marine plastic pollution crisis.

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